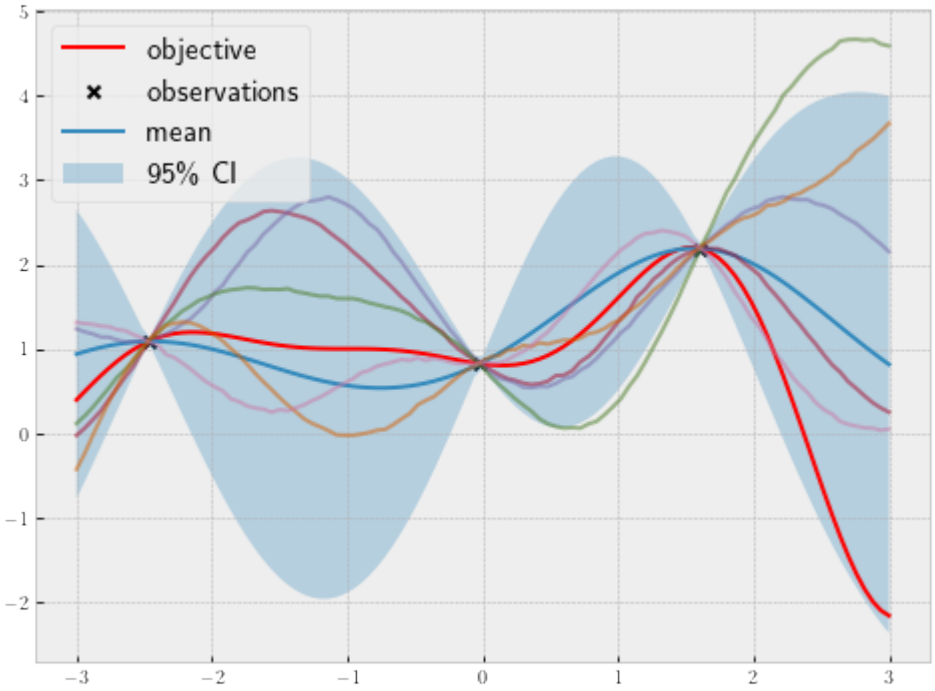


Figure 3.12 GP's predictions with the output scale set to 3. With a large output scale, the GP models the function to have a wider range, also allowing more uncertainty in the predictions.

While the left panel of Figure 3.11 and Figure 3.12 may look the same in that the GP and its samples have the same shape across the two plots, we notice that Figure 3.12 has a larger y-axis, as both its predictions and its samples take on larger values (both negative and positive). This is the direct result of scaling the covariance values from the RBF kernel with a large output scale.

With just two hyperparameters for our covariance function, we have seen that we can account for a wide range of functional behaviors that are modeled by our GP, which we summarize in Table 3.2.

Table 3.2 Summary of the roles that the length and output scales of a GP play.

Parameter	With a large value	With a small value
Length scale	Smoother predictions, less uncertainty	More variability, more uncertainty
Output scale	Larger output values, more uncertainty	Narrower output range, less uncertainty

I invite you to rerun the code we have with different values for the length and output scales to see their effects and verify Table 3.2 for yourself!

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